LESSON PLAN

QA/QC AND INDIGENISATION (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Effect of propeller rotation on aircraft and methods used to counteract it.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Effect of propeller rotation on aircraft and methods used to counteract it.
5. Specific Objectives	11.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of P-factor using a spinning propeller model; Safety: Ensure model is secured; Supervision: Instructor monitors student observation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) Define the term 'Gyroscopic Precession'.
(c) List three types of propellers based on pitch change mechanism.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Gyroscopic Precession Principles • Effect on Aircraft Control Surfaces • P-Factor Asymmetry Explanation • Torque Reaction on Fuselage • Propeller Slipstream Influence	L & Ds	PPT, IB	15 min
(b)	• Counteracting Torque Reaction • Managing P-Factor During Climb • Correcting Gyroscopic Effects	L & Ds	PPT, PDF	20 min
(c)	• Engine RPM and Propeller Speed • Aerodynamic Balance Techniques • Control Input Coordination	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how P-factor affects aircraft yaw during high angles of attack.
(b) Describe the specific control inputs required to counteract torque reaction.

12. Home Assignment and Follow Up	Prepare a brief note on how propeller rotation effects influence takeoff and landing performance, referencing the next lesson plan on engine power management.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min